SMARTBRIDGE
SKILLWALLET
A SmartBridge Product

Project Executable Files

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	3 Marks

Project Executable Files

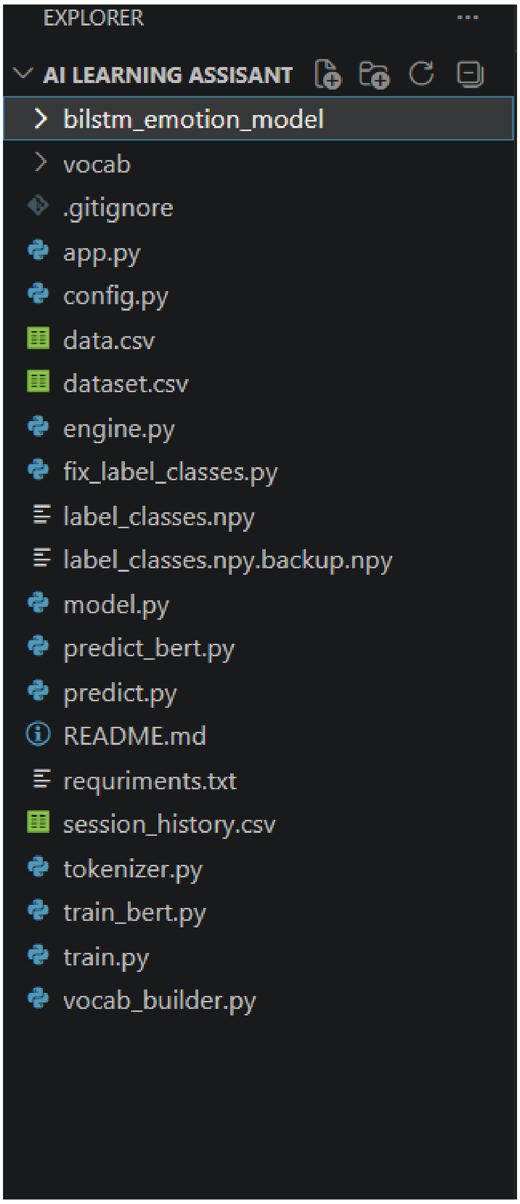
This section requires submission of all executable and deployable files for your project. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA/Partial)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	No — not yet written
3	requirements.txt / package.json / dependency file	Partial — requirement.txt exists but is missing tensorflow and torch
4	Database schema / seed files (if applicable)	NA — uses CSV file storage, no database
5	Environment configuration file (.env.example or similar)	Partial — a .env exists locally with a placeholder key, but no .env.example is committed
6	Deployed application URL (if hosted)	No — local host
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — web-based Streamlit app

8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	No — only ad-hoc __main__ test blocks exist in predictor.py and logger.py, no formal test suite
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure


EXPLORER
AI LEARNING ASSISANT
bilstm_emotion_model
vocab
.gitignore
app.py
config.py
data.csv
dataset.csv
engine.py
fix_label_classes.py
label_classes.npy
label_classes.npy.backup.npy
model.py
predict_bert.py
predict.py
README.md
requriments.txt
session_history.csv
tokenizer.py
train_bert.py
train.py
vocab_builder.py

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	
Login Credentials (Demo)	Not applicable — no login is required for the current prototype
Platform / Hosting Provider	Not yet hosted — planned: Streamlit Community Cloud
Repository Link	https://github.com/sivaadapa-ops/Emotion-Detection-and-Learning-Support-Engine/tree/main
Demo Video Link	https://drive.google.com/file/d/1og3Ph0JuiBqI3yKhtRCfNNcBlSrjqsW4/view?usp=sharing

Step 4: Run Instructions

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PROJECT EXECUTION / RUN INSTRUCTIONS

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Software Prerequisites

Before running the application, ensure the following software is installed on your system:

• Python 3.10 or later
• pip (Python Package Manager)
• Streamlit
• Visual Studio Code (Recommended)
• Git (Optional)

Step 1: Clone the Repository

Open a terminal or command prompt and execute:

https://github.com/sivaadapa-ops/Emotion-Detection-and-Learning-Support-Engine/tree/main

Navigate to the project folder:

cd AI-Learning-Assistant

Step 2: Create a Virtual Environment (Recommended)

Windows:

python -m venv venv
venv\Scripts\activate

Linux/macOS:

python3 -m venv venv
source venv/bin/activate

Step 3: Install the Required Dependencies

Install all the required Python packages using the following command:

pip install -r requirements.txt

Step 4: Verify Project Files

Before running the application, ensure the following files and folders are present in the project directory:

• app.py
• requirements.txt
• bilstm_emotion_model/
• vocab/
• dataset.csv
• data.csv
• label_classes.npy

These files are necessary for loading the trained models, vocabulary, datasets, and application components.

Step 5: Run the Streamlit Application

Execute the following command:

streamlit run app.py

Step 6: Open the Web Application

After executing the above command, Streamlit will automatically launch the application in your default web browser.

If it does not open automatically, manually visit:

http://localhost:8501

Step 7: Using the Application

1. Open the Streamlit web interface.
2. Enter a learning-related query or text input.
3. Submit the input using the available button.
4. The application processes the input using the trained NLP model.
5. The detected emotion and AI-generated response are displayed.
6. Conversation history is stored automatically (if enabled).

Step 8: Retraining the Models (Optional)

To retrain the BiLSTM model:

python train.py

To retrain the BERT model:

python train_bert.py

Step 9: Stop the Application

To stop the Streamlit server, return to the terminal window and press:

Ctrl + C

Expected Output

After successful execution:

• The Streamlit web application launches successfully.
• Users can interact with the AI Learning Assistant through the browser.
• The system predicts emotions from user input using deep learning models.
• Intelligent responses are generated based on the detected emotion.
• Conversation history is maintained for future interactions (if enabled).

Troubleshooting

If the application fails to start:

• Verify that Python is installed correctly.
• Ensure all dependencies are installed using:
  pip install -r requirements.txt

• Confirm that the required model folders
  (bilstm_emotion_model and vocab)
  are present.

• Verify that all dataset files are available in the project directory.

• If Streamlit is not installed, execute:

  pip install streamlit

• Restart the application using:

  streamlit run app.py

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End of Run Instructions

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Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	A keyword-based emotion override in app.py bypasses the real BiLSTM/BERT prediction for most inputs.	Needs to be removed so the actual model output drives the response.
2	bert_model.py's bert_predict() always returns a fixed "Joy" prediction regardless of input.	Needs to be replaced with real trained BERT inference.
3	logger.py contains two incompatible logging schemas writing to the same CSV file.	Needs consolidation into a single schema.